Chapter-13 Animation

1 CSS Animation

- Animation allows an element to change its styles automatically over time without using javaScript.

+ CSS Animation ki help se hum kisi box, button ya text ka color change kar sakte hai. use move kar sakte hai ya rotate kara sakte hai. Ye changes apne aap hota hai, bina javaScript ke.

2. Syntax

```
@keyframes animationName {
    from {
        /* Starting style */
    }

    to {
        /* Ending style */
    }
}

.element {
    animation: animationName 2s;
}
```

3 What is @keyframes?
- @keyframes defines the different stages of a CSS animation.

X @keyframes mein hum batate hai ki animation ke dauran element shuru mein kaisa hoga, beech mein kya change hoga aur end me kaisa dikhega.

Example 1 - Using from and to

```
<div class="box" ></div>                    ← HTML

.box {                                      ← CSS
    width: 100px;
    height: 100px;
    background: red;
→   animation: changeColor 3s infinite alternate;
}
@keyframes changeColor {
    from {
        background: red;
    }

    to {
        background: blue;
    }
}
```

Output:

Red box → Blue box → Red box.

Box starting me red hoga aur blue me change hoga. annant tak ye chalega. aur alternate use aage aur phir reverse direction me chalata hai.

Example 2- Using Percentage

```
.box {
    width : 100px;
    height : 100px;
    background: red;
    animation : colorChange 4s infinite;
}

@keyframes colorChange {
  0% {
    background : red;
  }
  50% {
    background : yellow;
  }
  100 % {
    background : blue;
  }
}
```

Output: Red → Yellow → Blue → Repeat

0% — animation ki shuruaat
50% — animation aadhi complete hone par
100% — animation ke end par.

Remember → from aur to starting aur ending styles batata hai. Aur percentages se hum beech ke animation bhi bata (defines) kar sakte hai.

4) Example – Change background Color

```
<div class="color-box"></div>
```
← HTML

```
.color-box {
  width : 100px;
  height : 100px;
  background : red;
→ animation : changColor 2s infinite alternate;
}

@keyframes changeColor {
  from {
    background : red;
  }
  to {
    background : blue;
  }
}
```

Output: Box ka color red se blue aur phir blue se red hota rahega.

DOMS | Page No.
Date / /

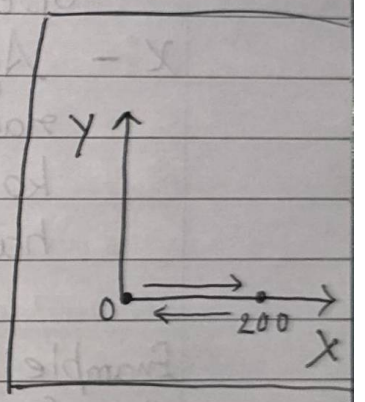
5. Example - Move a Box

```
<div class="moving-box"></div>

.moving-box {
  width : 50px;
  height : 50px;
  background : teal;
→ animation : moveBox 2s infinite alternate;
}

@keyframes moveBox {
  from {
    transform : translateX(0);
  }

  to {
    transform : translateX(200px);
  }
}
```



Output:
Box left se right move karega aur phir wapas aayega.

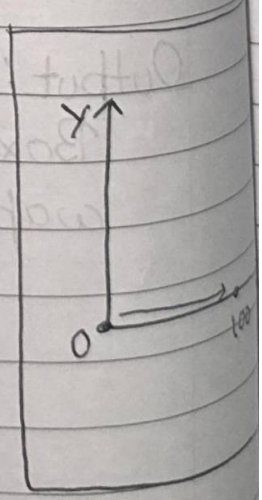
6. Important CSS Animation Properties.

i) animation-name
ii) animation-duration
iii) animation-delay
iv) animation-iteration-count
v) animation-direction
vi) animation-timing-function
vii) animation-fill-mode

i) animation-name
- Specifies the name of the animation.
x - Agar humne animation ka name moveBox rakha hai, toh is property se hum element ko batate hai ki usi animation ko chalana hai (moveBox animation ko chalana hai).

Example:

```
.box {
  animation-name : moveBox;
}
@keyframes moveBox {
  from {
    transform : translateX(0);
  }
  to {
    transform : translateX(100px);
  }
}
```



Output:
Box moveBox animation use karega.

(ii) animation-duration
- Defines how long one animation cycle takes.
X - Isse hum set karte hain ki animation ko ek bar me kitna time lagega, jaise 2 seconds ya 5 seconds.

Example:
```
.box {
  animation : moveBox 3s;
}
```

Output: Animation ka ek cycle 3 seconds mein complete hoga.

(iii) animation-delay
- Specifies how long to wait before the animation starts
X→Isse hum animation ko shuru karne se pehle wait karwa sakte hai. 2second dene par animation 2seconds baad shuru hogi.

Example:
```
.box {
  animation : moveBox 2s;
  animation-delay: 2s;
}
```

Output:
Animation shuru hone se pehle 2 seconds ka wait hoga, fhir uske baad start hoga.

iv) animation-iteration-count
- Defines how many times an animation should repeat.

X→ Isse hum decide karte hai ki animation kitne bar chalegi. 3 dene par 3 bar chalegi aur infinite dene par bar-bar chalti rahegi.

Example : 3 seconds

```
.box {
    animation : moveBox 2s;
    animation-iteration-count : 3s;
}
```

Output: Animation 3 baar chalegi.

Example : infinite

```
.box {
    animation-iteration-count
    animation : moveBox 3s infinite;
}
```

Output: Animation baar-baar repeat hoti rahegi.

v) animation-direction
- Specifies the direction in which an animation plays.

X → Ise ham decide ham karte hai ki animation normal direction mein chalegi, ulte direction mein chalegi ya aage jaane ke baad wapas bhi aayegi.

Example:
```
.box {
    animation : moveBox 2s infinite-alternate;
}
```

Output:
Box aage move karega aur phir wapas aayega.

Common value:
- normal → animation normal direction mein chalegi
- reverse → animation ulte direction mein chalegi
- alternate → pehle aage, phir wapas.
- alternate-reverse → pehle reverse direction, phir normal direction.

vi) animation-timing-function
- Controls how the speed of an animation changes over time.

X → Isse hum decide karte hain ki animation ek hi speed se chalegi ya shuru mein dheere, beech mein tez aur end mein phir dheere hogi.

Example:

```
.box {
    animation: moveBox 2s infinite;
    animation-timing-function : ease-in-out;
}
```

Output: Box ki movement shuru aur end mein dheemi hogi.

Common values:

- linear → Shuru se end tak ek hi speed.
- ease → shuru aur end mein dheemi, beech mein tez hoga.
- ease-in → dheere shuru hoti hai
- ease-out → End hote waqt dheemi hoti hai.
- ease-in-out → shuru aur end dono mein dheemi hoti hai.

(vii) animation-fill-mode

- Controls which animation styles apply before or after the animation runs.

X→ Isse hum decide karte hain ki animation khatam hone ke baad element apni aakhri animation wali style mein rahega ya nahi.

DOMS | Page No.
Date / /

Example:

```
.box {
  background : red;
  animation : changeColor 2s forwards;
}
@keyframes changeColor {
  from {
    background : red;
  }
  to {
    background : blue;
  }
}
```

Output: Animation khatam hone ke baad bhi box blue rahega.

Common values:

- none → animation ke baad final animated style retain nahi hoti.
- forwards → animation ke end wali style bani rehti hai.
- backwards → delay ke dauran starting keyframe style apply hoti hai.
- both → backl backwards aur forwards, dono ka behaviour apply hota hai.

DOMS | Page No.
Date / /

7) Shorthand Animation Property

- The animation shorthand lets us set multiple animation properties in one line.

X→ Har property ko alag-alag likhne ke bajaye, hum ek hi line mein animation ka naam, duration, speed, delay aur repeat set kar sakte hain.

Example:

```
.box {
  animation : moveBox 2s ease-in-out 1s infinite
              alternate;
}
```

Output:

Box 1 second baad move karna shuru karega aur aage-peeche chalta rahega.

Value	Meaning
moveBox	Animation ka naam
2s	Ek cycle ka time
ease-in-out	Shuru aur end mein dheemi speed
1s	Shuru hone se phale 1 second wait
infinite	Bar-Bar chalti rahegi
alternate	Age jayegi aur phir wapas aayegi.

Date / /

8 Animation Vs Transition

Animation	Transition
i) @keyframe use hota hai.	@keyframes ki zaroorat nahi.
ii) Ek animation mein kai step bana sakte hai	Aam taur par ek style se doosri style mein smooth changes hota hai.
iii) Apne aap shuru ho sakti hai	Aksar :hover ya kisi style change par shuru hoti hai.
iv) Baar-baar repeat kar sakte hain	iv) Apne aap baar-baar repeat nahi hoti.

→ Transition Example

```
<button> Hover Me </button>

button {
  background :blue;
  color : white;
  transition : background  0.5s;
}
button :hover {
  background : red;
}
```

Output:

Button par mouse le jaane par color blue se red change hoga.